CSCI-570 Homework 6

Due Date: Tuesday, November 25th, 11:59pm

Problem 1. [Think through]
Consider a flow network $G = (V, E)$ with a source $s \in V$, a sink $t \in V$, capacities $c : E \to \mathbb{N}^+$ and a flow f. Prove or disprove the following statements:

(a) If f is a max-flow, then $f(e) = 0$ or $f(e) = c(e)$ on all $e \in E$ edges.

(b) There is a max-flow for which $f(e) = 0$ or $f(e) = c(e)$ on all $e \in E$ edges.

(c) If all capacities are different, then there is a unique min-cut.

(d) Adding a constant number δ to all capacities $c(e)$ does not change the min-cuts.

(e) Multiplying all capacities $c(e)$ by a constant number δ does not change the min-cuts.

[5 points]

Problem 2.
Are max-flows and min-cuts unique?

(a) Give a simple example of a flow-network where the max-flow is unique, but it has several min-cuts.

(b) Give a simple example of a flow-network where the min-cut is unique, but is has several max-flows.

(c) Provide a polynomial-time algorithm which decides whether a flow-network has unique min-cut.

[Hint for (c): what can happen to the value of the max-flow if you change the capacities of a min-cut?]

[2+2+6 points]

Problem 3.
Let $G = (V, E)$ be a flow network with source s, sink t, and integer capacities. Suppose that we are given a maximum flow f in G.

(a) Suppose that we *increase* the capacity of a single edge $(u, v) \in E$ by 1.
Give an $O(n + m)$ (i.e. $O(V + E)$) runtime algorithm to update the maximum flow.

(b) Suppose that we *decrease* the capacity of a single edge $(u, v) \in E$ by 1.
Give an $O(n + m)$ (i.e. $O(V + E)$) runtime algorithm to update the maximum flow.

[Hint: one iteration of the Ford-Fulkerson algorithm and graph traversal algorithms each can be run in $O(n + m)$.]

[10 points]

Problem 4.
We define the *Escape Problem* as follows. We are given a directed graph $G = (V, E)$ (imagine a network of roads). A certain collection of nodes $X \subset V$ are designated as populated nodes, and a certain other collection $S \subset V$ are designated as safe nodes. (Assume that X and S are disjoint.) In case of an emergency, we want evacuation routes from the populated nodes to the safe nodes. A set of evacuation routes is defined as a set of paths in G so that (i) each node in X is the tail of one path, (ii) the last node on each path lies in S, and (iii) the paths do not share any edges. Such a set of paths gives a way for the occupants of the populated nodes to "escape" to S, without overly congesting any edge in G.

(a) Given G, X, and S, show how to decide in polynomial time whether such a set of evacuation routes exists.

(b) Suppose we have exactly the same problem as in (a), but we want to enforce an even stronger version of the "no congestion" condition (iii). Thus we change (iii) to say "the paths do not share any nodes." Provide a polynomial time algorithm to decide whether such a set of evacuation routes exists.

[Hint for (b): split vertices v into v_{in} and v_{out} and connect them with a unit capacity: $v_{in} \xrightarrow{1} v_{out}$]

[3+5 points]